

Assignment 1

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Part 1)

Objective Function

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

$$f(x) = x^2 + 5, \quad f'(x) = 2x, \quad f''(x) = 2$$

$y_0 = 0$ for all five points

$$\hookrightarrow D(x) = (x - x_0)^2 + (x^2 + 5 - 0)^2 = (x - x_0)^2 + (x^2 + 5)^2$$

$$D'(x) = 2(x - x_0) + 2(x^2 + 5)(2x) = 4x^3 + 22x - 2x_0 = 2(2x^3 + 11x - x_0)$$

$$D''(x) = 12x^2 + 22 = 2(6x^2 + 11) > 0 \text{ for all } x \in \mathbb{R}, \text{ one global min for each point}$$

Analytical Approach

$$\text{Set } D'(x) = 0$$

$$2(2x^3 + 11x - x_0) = 0$$

$$2x^3 + 11x - x_0 = 0$$

$$\text{Point } (0, 0): 2x^3 + 11x = 0$$

$$x(2x^2 + 11) = 0$$

$$x = 0 \quad \swarrow \quad \searrow \quad 2x^2 + 11 = 0 \text{ (no sol'n)}$$

$$\therefore x = 0$$

$$f(x) = 0^2 + 5 = 5$$

Closest point is $(0, 5)$

$$d = \sqrt{(x - x_0)^2 + (x^2 + 5)^2}$$

$$d = \sqrt{0^2 + 5^2} = 5$$

$$\text{Point } (-4, 0): 2x^3 + 11x + 4 = 0$$

$$\therefore x = -0.355$$

$$f(x) = (-0.355)^2 + 5 = 5.126$$

Closest point is $(-0.355, 5.126)$

$$d = \sqrt{(-0.355 - (-4))^2 + (-0.355^2 + 5)^2} = 6.290$$

$$\text{Point } (-8, 0): 2x^3 + 11x + 8 = 0$$

- 1.77

Point $(-8, 0)$: $2x^3 + 11x + 8 = 0$

$\therefore x = -0.672$

$f(x) = (-0.672)^2 + 5 = 5.452$

Closest point is $(-0.672, 5.452)$

$d = \sqrt{(-0.672 - (-8))^2 + (-0.672^2 + 5)^2} = 9.133$

Point $(2, 0)$: $2x^3 + 11x - 2 = 0$

$\therefore x = 0.181$

$f(x) = (0.181)^2 + 5 = 5.033$

Closest point is $(0.181, 5.033)$

$d = \sqrt{(0.181 - 2)^2 + (0.181^2 + 5)^2} = 5.351$

Point $(6, 0)$: $2x^3 + 11x - 6 = 0$

$\therefore x = 0.520$

$f(x) = (0.520)^2 + 5 = 5.270$

Closest point is $(0.520, 5.270)$

$d = \sqrt{(0.520 - 6)^2 + (0.520^2 + 5)^2} = 7.603$

Numerical Approach

Approach 1: Newton-Raphson

Set $g(x) = 0$:

$x_{n+1} = x_n + \Delta x_n, \quad \Delta x_n = -H_n^{-1} g_n = -g(x_n)/H(x_n)$

We step size $\delta_n = 1$:

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

For each point, start with $x_0 = 0$, and stop when $|x_{n+1} - x_n| < \text{tol} = 10^{-3}$

Point $(0, 0)$: $n = 0, \quad x_0 = 0$

$g(x_0) = D'(0) = 4(0)^3 + 22(0) - 2(0) = 0$

$H(x_0) = D''(0) = 12(0)^2 + 22 = 22$

$\Delta x_0 = 0/22 = 0 \quad x_{n+1} = 0$

Since $|x_{n+1} - x_n| = 0 < 10^{-3}$, converges

$\therefore x = 0, \quad d = 5$

Point $(-4, 0)$:

n	x_n	$g(x_n)$	$H(x_n)$	Δx_n	x_{n+1}
0	0	8	22	-3.64×10^{-1}	-3.64×10^{-1}
1	-3.64×10^{-1}	-0.2	23.6	8.47×10^{-3}	-3.56×10^{-1}
2	-3.56×10^{-1}	-0.01	23.5	4.26×10^{-4}	-3.56×10^{-1}

$\therefore x = -3.56 \times 10^{-1}$, $d = 6.290$

Point $(-6, 0)$:

n	x_n	$g(x_n)$	$H(x_n)$	Δx_n	x_{n+1}
0	0	16	22	-7.27×10^{-1}	-7.27×10^{-1}
1	-7.27×10^{-1}	-1.5	28.4	5.28×10^{-2}	-6.74×10^{-1}
2	-6.74×10^{-1}	0	27.4	0	-6.74×10^{-1}

$\therefore x = -6.74 \times 10^{-1}$, $d = 9.133$

Point $(2, 0)$:

n	x_n	$g(x_n)$	$H(x_n)$	Δx_n	x_{n+1}
0	0	-4	22	1.82×10^{-1}	1.82×10^{-1}
1	1.82×10^{-1}	0.02	22.4	-8.93×10^{-4}	1.81×10^{-1}
2	1.81×10^{-1}	0	22.4	0	1.81×10^{-1}

$\therefore x = 1.81 \times 10^{-1}$, $d = 5.351$

Point $(6, 0)$:

n	x_n	$g(x_n)$	$H(x_n)$	Δx_n	x_{n+1}
0	0	-12	22	5.45×10^{-1}	5.45×10^{-1}
1	5.45×10^{-1}	0.6	25.6	-2.34×10^{-2}	5.22×10^{-1}
2	5.22×10^{-1}	0.1	25.3	-3.95×10^{-3}	5.18×10^{-1}
3	5.18×10^{-1}	0	25.2	0	5.18×10^{-1}

Approach 2: Golden Section Search

Use $[a, b] = [-2, 2]$ $\phi = (1 + \sqrt{5})/2 = 1.618$ $r = 2 - \phi = 0.382$

$$x_1 = a + r(b-a) \quad x_2 = b - r(b-a) \quad \text{tolerance} = 10^{-7}$$

$$\text{iterations} = \lceil \ln(4/10^{-7}) / \ln \phi \rceil = 37$$

If $D(x_1) < D(x_2)$, minimum can't be in $(x_2, b]$, so set $x_2 \rightarrow b$, otherwise $x_1 \rightarrow a$.

Point $(0, 0)$: $D(x) = (x-0)^2 + (x^2+5-0)^2 = x^2 + (x^2+5)^2$

$n=0$: $x_1 = -0.472$ $x_2 = 0.472$ $D(x_1) = 27.5$ $D(x_2) = 27.5$

Point (0,0): $D(x) = (x-0)^+ + (x^- + 5 - 0)^+ = x^2 + (x^- + 5)^+$

$h=0$: $x_1 = -0.472$ $x_2 = 0.472$ $D(x_1) = 27.5$ $D(x_2) = 27.5$

$D(x_1) < D(x_2)$ is false. keep $a = x_1 = -0.472$

$x_1 = \text{old } x_2 = 4.72 \times 10^{-1}$ $D(x_1) = 27.5$

$x_2 = 2 - 0.382 (2 - (-0.472)) = 1.06$

Iteration done in excel file (golden-section-search-iteration.xlsx)

$x = 0$ $d = 5$ $\hookrightarrow$ for all points

Point (-4,0):	n	a	b	x_1	x_2	$D(x_1)$	$D(x_2)$	keep
	0	-2	2	-0.472	0.472	39.7	47.3	$b = x_2$

$\downarrow \dots$

$x = -0.355$ $d = 6.290$

Point (-8,0): $x = -0.672$ $d = 9.133$

Point (2,0): $x = 0.181$ $d = 5.351$

Point (6,0): $x = 0.520$ $d = 7.603$

Part 2)

Set-up

Data $(x_i, y_i) = (0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)$

$\hookrightarrow N = 4$

$$MSE = \frac{1}{N} \sum_{i=1}^N r_i^2 = \frac{1}{N} \|Ax - b\|_2^2 = \frac{RSS}{N}$$

$\sum x_i = 6$ $\sum x_i^2 = 14$ $\sum x_i^3 = 36$ $\sum x_i^4 = 98$

$\sum y_i = 13$ $\sum x_i y_i = 31$ $\sum x_i^2 y_i = 83$ $\sum y_i^2 = 71$

Line $(y^*(x) = c_0 + c_1 x)$

$$MSE(c_0, c_1) = \frac{1}{4} \sum_{i=1}^4 (c_0 + c_1 x_i - y_i)^2 = c_0^2 + 3c_0 c_1 + 3.5c_1^2 - 6.5c_0 - 15.5c_1 + 17.75$$

Analytical Solution

$$(A, x_n) \begin{pmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \approx \begin{pmatrix} 0.5 \\ 3.5 \\ 1.5 \\ 7.5 \end{pmatrix}$$

$$(A \approx y) \begin{pmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \approx \begin{pmatrix} 0.5 \\ 3.5 \\ 1.5 \\ 7.5 \end{pmatrix}$$

$$X^T A \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} 13 \\ 31 \end{pmatrix}$$

$$c_0 = -0.2 \quad c_1 = 2.3$$

$$y^*(x) = -0.2 + 2.3x$$

$$\therefore \text{MSE}(c_0, c_1) = 0.575$$

Numerical Solution (Newton-Raphson)

$$g_{c_0} = \frac{\partial \text{MSE}}{\partial c_0} = 2c_0 + 3c_1 - 6.5 \quad H_{c_0 c_0} = \frac{\partial^2 \text{MSE}}{\partial c_0^2} = 2$$

$$g_{c_1} = \frac{\partial \text{MSE}}{\partial c_1} = 3c_0 + 7c_1 - 15.5 \quad H_{c_1 c_1} = \frac{\partial^2 \text{MSE}}{\partial c_1^2} = 7$$

$$c_0(\text{cnt}+1) = c_0(\text{cnt}) - \frac{g_{c_0}}{H_{c_0 c_0}} = \frac{6.5 - 3c_1(\text{cnt})}{2}$$

$$c_1(\text{cnt}+1) = c_1(\text{cnt}) - \frac{g_{c_1}}{H_{c_1 c_1}} = \frac{15.5 - 3c_0(\text{cnt})}{7}$$

$$\text{Initial guess: } c_0(0) = 0, \quad c_1(0) = 0 \quad \text{MSE}_0 = 17.8$$

$$n=1: c_1(1) \approx 2.21 \quad c_0(1) = -0.071 \quad \text{MSE}_1 = 0.584$$

→ Repeated in excel until convergence (mse_iteration.xlsx)

$$\therefore n=32: c_0 = -0.2 \quad c_1 = 2.3 \quad \text{MSE} \approx 0.575 \quad \text{tolerance} = 10^{-7}$$

Parabola ($y = Bx^2 + Cx + D$)

Set-up

$$\text{MSE}(B, C, D) = \frac{1}{4} \sum_{i=1}^4 (Bx_i^2 + Cx_i + D - y_i)^2$$

$$= \frac{1}{4} [(D - 0.5)^2 + (4B + 2C + D - 3.5)^2 + (B + C + D - 1.5)^2 + (9B + 3C + D - 7.5)^2]$$

Analytical Solution

$$X^T A \begin{pmatrix} 0 & 0 & 1 \\ 4 & 2 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix} \begin{pmatrix} B \\ C \\ D \end{pmatrix} \approx \begin{pmatrix} 0.5 \\ 3.5 \\ 1.5 \\ 7.5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 9 & 36 & 14 \end{pmatrix} \begin{pmatrix} B \\ C \\ D \end{pmatrix} = \begin{pmatrix} 83 \end{pmatrix}$$

$$X^T \begin{pmatrix} 9 & 3 & 1 \end{pmatrix} \begin{pmatrix} 7.5 \\ 83 \\ 31 \\ 13 \end{pmatrix}$$

$$\begin{pmatrix} 98 & 36 & 14 \\ 36 & 14 & 6 \\ 14 & 6 & 4 \end{pmatrix} \begin{pmatrix} B \\ C \\ D \end{pmatrix} = \begin{pmatrix} 83 \\ 31 \\ 13 \end{pmatrix}$$

$$D = 0.55 \quad C = 0.05 \quad B = 0.75$$

$$y = 0.75x^2 + 0.05x + 0.55$$

$$\therefore \text{MSE} = 0.0125$$

Numerical Solution (Newton-Raphson)

$$g_B = 49B + 18C + 7D - 41.5 \quad H_{BB} = 49$$

$$g_C = 18B + 7C + 3D - 15.5 \quad H_{CC} = 7$$

$$g_D = 7B + 3C + 2D - 6.5 \quad H_{DD} = 2$$

$$B_{n+1} = \frac{41.5 - 18C_n - 7D_n}{49}$$

$$C_{n+1} = \frac{15.5 - 18B_{n+1} - 3D_n}{7}$$

$$D_{n+1} = \frac{6.5 - 7B_{n+1} - 3C_{n+1}}{2}$$

$$\text{Initial guess: } B_0 = C_0 = D_0 = 0 \quad \text{MSE}_0 = 17.8$$

$$n=1: B_1 = 0.847 \quad C_1 = 0.036 \quad D_1 = 0.231 \quad \text{MSE}_1 = 0.118$$

→ repeated in excel until convergence (mse-iteration.xlsx)

$$\therefore n=276: B = 0.75 \quad C = 0.05 \quad D = 0.55 \quad \text{MSE} = 0.0125 \quad \text{tolerance} = 10^{-7}$$